One-page project proposal

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Question: what do weather and regional fire histories add beyond PM history, and which 3/5/7-day window is useful?

Issue and target: 07:00 Thailand on D -> modeled daily CAMS profile P(D+1); 17h to target start. One Chiang Mai receptor. Study target dates 2024-2025; source buffers precede the period.

Sources: fixed 12 UTC CAMS forecasts, Open-Meteo GFS previous_day1 weather forecasts and GFAS v1.2 UTC daily analysis. Air4Thai/Open-Meteo provides a separate contemporary C6 reference, not historical target truth.

Design: H1 PM, H2 PM+weather, H3 PM+fire, H4 all, each at 3/5/7 days on identical common dates. Source-specific clocks, 24-hour completeness, no target/predictor imputation, conservative advisory dependencies and mature-label boundary purges.

Evaluation: persistence and direct CAMS references; primary Ridge, secondary HGB; training-only temporal CV and validation-only source/window choice. Full calendar 2025 test, MAE/RMSE and fixed 37.5 warning metrics; exploratory calendar-block intervals.

Coverage: 731 study target days,704 value-complete source histories,687 common modeling dates after declared quality/maturity rules. Technical feasibility does not certify original public release/revision.

Outputs: four executed notebooks, required scripts, exact raw/processed data, SQLite audit store, figures, metrics, development comparison and the English lab report.

Scope: retrospective benchmark after prior exploration. Temporal cutoff, label maturity and training-only preprocessing address Section 6.2; exact historical release/revision and deployment are separate, unverified claims. Student review, repository access and video submission remain required.